Lab session 3: Bipolar-Transistor

1. OBJECTIVE OF THIS LAB SESSION

In this lab, various characteristic curves of bipolar transistors and an emitter circuit are to be investigated. The BC546A small signal transistor is used for all subsequent parts.

Part 1: Output characteristics

Theory

The family of output characteristics $I_C = f(U_{CE})$ with $I_B = const.$ of the transistor BC546A is to be measured with the X-Y-recorder or with the oscilloscope. The base current is to be fed using a voltage source.

Preparation 1

- Find the dimensions & important characteristic values for the transistor in the corresponding data-sheet.
- The collector current shall be varied in a range of $I_C = 0 \cdots 2mA$. Calculate the corresponding base currents I_B and find a possibility to set these currents by a voltage source that is adjustable in the range of $0 \ldots 20V$.
- Draw a circuit diagram to plot the transistor current I_C on the Y-axis and the collector emitter voltage U_{CE} on the X-axis of a X-Y recorder or using the oscilloscope.

Lab-Task 1:

- Record the family of output characteristics for the range $I_C = 0 \cdots 2mA$ and $U_{CE} = 0 \cdots 10V$ for 5 different base currents. Remember to store the scaling of the recorder to later analyze the recorded curves.

Evaluation 1

- Determine the current amplification at $U_{CE} = 5V$ and display it graphically in the dependency of the collector current. Calculate the average current gain value (SPICE-parameter bf).
- Graphically determine the “early voltage” (SPICE-parameter vaf).

Part 2: Transfer characteristics

Theory

The transfer characteristics $I_C = f(U_{BE})$ with $U_{CE} = const.$ of the transistor BC546A is to be measured with the preferred recorder.

Preparation 2

- Draw a circuit diagram to plot the transistor current I_C on the Y-axis and the base-emitter voltage U_{BE} on the X-axis.

Lab-Task 2:

- Record the transfer characteristics for the range $I_C = 0 \cdots 2mA$ and $U_{CE} = 5V$. Remember to store the scaling of the recorder to later analyze the recorded curves.

Evaluation 2

- Graphically determine the transconductance for $I_C = 1mA$.
- Recall you finding of lab 2 (Diode) to determine the saturation current I_S under the assumption $m = 1$.
- Use the parameters I_S, U_A, and B_f of the transistor to simulate the family of output characteristics and transmission characteristic for the specified value ranges with a SPICE program, and compare the results with your measurements

Part 3: Common-Emitter Circuit

Theory

In the following task the common-emitter circuit as illustrated in the figure below shall be investigated. The DC-source U_B is used to set-up the operating point, the AC-source U_1 is acting as a small signal that shall be amplified to a voltage U_2.

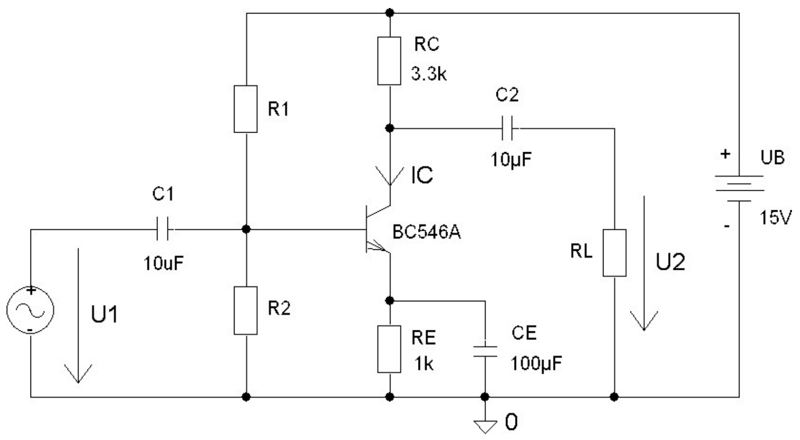

Figure: Common-Emitter Circuit

Preparation 3

- The voltage divider at the transistor base connector shall be dimensioned for a collector current of $I_C = 2mA$. Remark: In the laboratory there are resistors of the E12 series – use resistor values that match best.
- Use the resistance values used to calculate the operating point, AC-voltage gain and the input and output resistance

Lab-Task 3

- Set-up the circuit as shown above
- Measure the operating point of the circuit and the AC-voltage gain for a for AC-signal of $f = 1kHz$, and $U_1 = 10mV$
- Measure the input and output resistance according to the half-voltage method for the **output** voltage.

Hint: Connect a resistor decade as load respectively in series to C1 and vary it until the voltage dropped to half of its open circuit voltage. Please ensure that the Scope is in AC mode for this measurement.

Evaluation 3

- Compare the theoretic values with the measured results.